

Notes: Heston & Rouwenhorst (JFE, 1994)

Does Industrial Structure Explain the Benefits of International Diversification?

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1 Overview

International diversification works in Europe not mainly because countries hold different mixes of industries, but because there are *country-specific return components* that are large relative to industry-specific components. Even after “control” for industry composition, country indices remain weakly correlated. So the gains from holding many countries are not just disguised industry diversification (and they are not mainly currency diversification either).

2 Research question and empirical setting

- **Question:** Are low cross-country correlations (and hence big gains from international diversification) just the mechanical result of different industrial structures across countries?
- **Idea:** A country index is a weighted average of industry returns. If countries differ mostly in industry weights, then “country diversification” is basically “industry diversification.”
- **Data:** MSCI constituents/indices for (mainly) Europe. Monthly stock returns, 12 countries, 7 broad industries, sample period 1978–1992. Returns are studied for both equally-weighted (EW) and value-weighted (VW) portfolios.

3 Model 1: Two-way (country $\times$ industry) return decomposition

3.1 Notation

- $i \in \{1, \dots, N\}$ indexes stocks.
- $k(i) \in \{1, \dots, K\}$ is the country of stock i ($K = 12$).
- $j(i) \in \{1, \dots, J\}$ is the industry of stock i ($J = 7$).
- $R_{i,t}$ is the return of stock i in month t (measured in a common currency in the baseline).

3.2 Additive factor structure (monthly cross-sectional ANOVA)

Each month t , the paper decomposes returns into a European market component, a country component, and an industry component:

$$R_{i,t} = \alpha_t + \beta_{j(i),t} + \gamma_{k(i),t} + \varepsilon_{i,t}. \quad (1)$$

- α_t : the “European market” return.
- $\beta_{j,t}$: the *industry* effect in month t .
- $\gamma_{k,t}$: the *country* effect in month t .
- $\varepsilon_{i,t}$: stock-specific residual.

Think of (1) as a two-way fixed-effect model run separately each month.

3.3 Dummy-variable regression form

Let $I_{i,j}$ be the industry dummy and $C_{i,k}$ the country dummy:

$$I_{i,j} = \mathbf{1}\{j(i) = j\}, \quad C_{i,k} = \mathbf{1}\{k(i) = k\}. \quad (2)$$

Then (1) can be written as:

$$R_{i,t} = \alpha_t + \sum_{j=1}^J \beta_{j,t} I_{i,j} + \sum_{k=1}^K \gamma_{k,t} C_{i,k} + \varepsilon_{i,t}. \quad (3)$$

Because there is a constant plus a full set of country and industry dummies, identification requires normalizations.

3.4 Identification restrictions (EW OLS)

For equally-weighted estimation (OLS), the paper uses count-weighted zero-sum restrictions:

$$\sum_{j=1}^J n_j \beta_{j,t} = 0, \quad (4)$$

$$\sum_{k=1}^K m_k \gamma_{k,t} = 0, \quad (5)$$

where n_j is the number of stocks in industry j (Europe-wide) and m_k is the number of stocks in country k .

Interpretation: (4) pins down α_t as the return on the *European equally-weighted market*. Similarly, $\gamma_{k,t}$ becomes a “pure” country deviation from the European market (holding the European industry mix fixed).

3.5 Identification restrictions (VW WLS)

For value-weighted estimation (WLS), the paper uses market-value weights:

$$\sum_{j=1}^J w_j \beta_{j,t} = 0, \quad (6)$$

$$\sum_{k=1}^K v_k \gamma_{k,t} = 0, \quad (7)$$

where w_j is industry j ’s weight in the European VW market and v_k is country k ’s weight in the European VW market.

3.6 Economic meaning of the components

The restrictions imply clean portfolio interpretations:

- α_t is the return on the European market (EW or VW depending on estimation).
- $\alpha_t + \beta_{j,t}$ is the return on a *Europe-wide industry portfolio* (geographically diversified within industry j).
- $\alpha_t + \gamma_{k,t}$ is the return on a *pure country portfolio* (industry-diversified within country k using European industry weights).

4 Model 2: Mapping stock-level effects into index returns

Let $R_{k,t}^{\text{EW}}$ be the EW return on the country- k index (average of stocks in k). Averaging (1) over stocks in country k yields:

$$R_{k,t}^{\text{EW}} = \alpha_t + \sum_{j=1}^J \omega_{k,j}^{\text{EW}} \beta_{j,t} + \gamma_{k,t}, \quad (8)$$

where $\omega_{k,j}^{\text{EW}}$ is the share of country- k stocks in industry j (counts-based).

Similarly, for an EW industry index $R_{j,t}^{\text{EW}}$:

$$R_{j,t}^{\text{EW}} = \alpha_t + \beta_{j,t} + \sum_{k=1}^K \omega_{j,k}^{\text{EW}} \gamma_{k,t}, \quad (9)$$

where $\omega_{j,k}^{\text{EW}}$ is the share of industry- j stocks located in country k .

4.1 “Corrected” indices used in Table 4

The paper then constructs:

- **Country index corrected for industry effects:**

$$\tilde{R}_{k,t} \equiv \alpha_t + \gamma_{k,t}. \quad (10)$$

This removes the mechanical effect of a country’s industrial mix.

- **Industry index corrected for country effects:**

$$\tilde{R}_{j,t} \equiv \alpha_t + \beta_{j,t}. \quad (11)$$

This removes the mechanical effect of an industry’s country composition.

5 Tables

5.1 Table 1: Industry composition differs across countries (motivation)

What we read from it:

- Countries look quite different in sectoral composition (counts and value weights). This is the core motivation for the “industrial structure” hypothesis.
- But even large sector-weight differences are not automatically enough to explain low cross-country correlations if (i) industries are highly correlated with each other and/or (ii) there are big country shocks.

Table 1
Industry composition by countries
Industry composition of the Morgan Stanley Capital International country indices 1970–1992.
Panel A gives for each country by industry the number of firms included in the sample. Panel B gives the average market capitalization of the countries by industry expressed as a percentage of the European value-weighted market.

Country	Industry							
	F	E	U	T	Co	Ca	B	Total
A: Number of firms by country and industry								
Austria	7	1	1	1	3	2	9	24
Belgium	9	1	8	1	3	—	16	38
Denmark	10	—	1	5	14	3	2	35
France	29	4	2	1	46	17	16	115
Germany	18	1	3	3	22	17	27	91
Italy	28	3	5	2	19	11	19	85
Netherlands	7	1	—	3	10	4	7	32
Norway	9	1	—	5	10	9	6	40
Spain	11	2	9	2	11	2	14	52
Sweden	5	—	—	—	13	11	26	55
Switzerland	19	—	3	3	11	11	9	72
United Kingdom	38	9	18	4	49	36	35	189
Europe	189	23	47	30	231	123	186	829
B: Average weights in the European value-weighted market								
Austria	0.13	0.02	0.01	0.01	0.08	0.03	0.10	0.38
Belgium	0.09	0.01	0.74	0.02	0.14	0.31	2.71	
Denmark	0.30	—	0.01	0.13	0.43	0.07	0.01	0.95
France	1.86	1.14	0.42	0.03	3.77	1.39	1.30	10.11
Germany	5.58	0.10	0.92	0.26	3.89	3.05	5.37	19.17
Italy	2.65	0.07	0.99	0.03	1.18	0.44	0.46	5.23
Netherlands	0.74	3.56	—	0.17	1.39	0.71	0.37	6.94
Norway	0.14	0.36	—	0.04	0.12	0.12	0.05	0.83
Spain	1.47	0.18	1.85	0.03	0.35	0.01	0.22	4.02
Sweden	0.28	—	—	—	0.83	0.72	0.64	2.47
Switzerland	1.98	—	—	—	2.84	0.35	1.08	8.41
United Kingdom	9.06	6.26	2.40	0.43	12.04	4.65	3.96	38.79
Europe	26.87	12.50	6.75	1.30	26.97	11.53	14.07	100.00

F = Finance, Insurance, and Real Estate, E = Energy, U = Utilities, T = Transportation and Storage, Co = Consumer Goods and Services, Ca = Capital Goods, B = Basic Industries.

Table 2
Summary statistics (monthly data 1970–1992)

Panel A summarizes the mean and the standard deviation of the monthly equally-weighted (EW) and value-weighted (VW) country index returns. Panel B correlates the country returns for the monthly returns on equally- and value-weighted industry portfolios. All returns are measured in US dollars and reported in percent per month. The country return is the proportional change in the exchange rate measured in DM per unit of local currency. The correlations above the diagonal refer to value-weighted index returns, those below the diagonal are between the equally-weighted index returns.

A: By country	EW return	VW return	Currency return	Correlation matrix																	
Country	Mean	Std. dev.	Mean	Std. dev.	A	B	D	F	G	I	N	Sp	Sw	UK	Eu						
Austria	(A)	0.795	6.580	0.646	4.617	0.777	0.543	0.095	0.333	0.234	0.234	0.287	0.282	0.289	0.274	0.480	0.247	0.663			
Belgium	(B)	1.220	5.717	1.156	5.082	−0.149	0.827	0.386	0.537	0.328	0.524	0.273	0.275	0.257	0.289	0.612	0.325	0.471	0.666		
Denmark	(D)	1.151	5.480	1.051	5.536	−0.382	1.267	0.134	0.341	—	0.239	0.130	0.241	0.411	0.233	0.527	0.374	0.370	0.242		
France	(F)	1.489	6.489	1.386	6.486	−0.228	1.060	0.307	0.571	0.266	—	0.051	0.086	0.475	0.440	0.366	0.405	0.414	0.645		
Germany	(G)	0.852	4.718	0.884	5.361	0.021	0.611	0.366	0.332	0.362	0.346	0.262	0.337	0.400	0.440	0.391	0.391	0.719			
Italy	(I)	1.264	7.123	1.356	7.062	−0.627	1.593	0.236	0.583	0.327	0.417	0.443	0.386	0.246	0.227	0.271	0.333	0.550			
Netherlands	(N)	1.118	5.513	1.101	4.902	−0.019	0.421	0.337	0.306	0.448	0.440	0.331	0.267	0.378	0.471	0.421	0.400	0.707			
Norway	(N)	0.578	7.844	1.047	8.522	−0.284	1.798	0.334	0.347	0.445	0.385	0.444	0.413	0.463	0.378	0.410	0.467	0.591	0.385		
Spain	(Sp)	1.240	6.260	1.060	6.508	0.321	1.162	0.238	0.447	0.352	0.447	0.447	0.396	0.440	0.412	0.491	0.449	0.554			
Sweden	(Sw)	1.614	7.461	1.367	7.074	−0.351	2.122	0.241	0.387	0.322	0.193	0.243	0.386	0.477	0.476	0.440	0.476	0.472	0.693		
Switzerland	(Sw)	0.719	4.937	0.915	4.732	−0.041	1.436	0.416	0.460	0.400	0.479	0.493	0.582	0.673	0.497	0.644	0.497	0.671	0.735		
United Kingdom	(UK)	1.414	6.364	1.356	6.013	−0.328	2.047	0.313	0.387	0.423	0.471	0.461	0.507	0.507	0.481	0.479	0.570	0.878			
Europe	(Eu)	1.234	4.524	1.107	4.412	0.580	0.741	0.553	0.343	0.740	0.645	0.711	0.622	0.681	0.619	0.570	0.678	0.931	0.881		

B: By industry sector	EW return		VW return		Correlation matrix												
	Mean	Std. dev.	Mean	Std. dev.	F	E	U	T	Co	Ca	B	Eu					
Finance	(F)	1.085	4.414	1.085	4.412	0.517	0.678	0.767	0.887	0.887	0.887	0.887					
Energy	(E)	1.479	5.911	1.479	5.910	0.626	0.628	0.493	0.292	0.568	0.555	0.715					
Utilities	(U)	1.071	3.961	0.995	4.423	0.710	0.446	0.614	0.626	0.649	0.646	0.710					
Transport	(T)	1.081	5.104	0.846	5.397	0.008	0.535	0.621	0.775	0.788	0.816	0.884					
Consumer	(Co)	1.487	4.331	1.297	4.603	0.055	0.669	0.786	0.786	0.913	0.899	0.934					
Capital	(Ca)	0.874	5.220	0.864	5.584	0.015	0.447	0.711	0.819	0.816	0.903	0.934					
Basic	(B)	1.181	5.101	0.963	4.777	0.030	0.639	0.786	0.836	0.880	0.901	0.937					
Europe	(Eu)	1.234	4.524	1.105	4.412	0.668	0.684	0.710	0.646	0.678	0.694	0.931					

5.2 Table 2: Raw index moments — countries are weakly correlated, industries strongly correlated

Panel A (by country):

- Country indices have substantial volatility, and the country-by-country correlation matrix shows only moderate co-movement.
- This is the empirical reason international diversification looks attractive.

Panel B (by industry):

- Industry indices are much more correlated with one another than country indices are with each other.
- Intuition: within Europe, industries share a large common component (the European market cycle).

Key comparison (takeaway): If the only reason countries differ is that they load differently on a small set of industry factors, I would expect country correlations to be closer to industry correlations. Table 2 suggests that something beyond industrial mix is operating.

5.3 Table 3: Variance decomposition — country effects dominate excess country returns

Table 3 is the main punchline. Define the excess country return over the European market:

$$R_{k,t}^{\text{excess}} \equiv R_{k,t} - \alpha_t = \gamma_{k,t} + \sum_{j=1}^J \omega_{k,j} \beta_{j,t}. \quad (12)$$

Then

$$\text{Var}(R_k^{\text{excess}}) = \text{Var}(\gamma_k) + \text{Var}\left(\sum_j \omega_{k,j} \beta_j\right) + 2 \text{Cov}\left(\gamma_k, \sum_j \omega_{k,j} \beta_j\right). \quad (13)$$

What Table 3 reports: the time-series variance of the “pure country” component $\gamma_{k,t}$ and the variance of the “industry-mix” component $\sum_j \omega_{k,j} \beta_{j,t}$, each scaled by the variance of the full excess return.

Interpretation:

- For EW country indices, the variance share of the industry-mix term is essentially zero on average (well below 1%).
- For VW country indices, the industry-mix term is still small on average (single-digit percent range).
- Some variance shares exceed 1 because the covariance term can be negative (so component variances need not add up exactly).

Industry-side panel: Excess industry returns over the European market decompose as

$$R_{j,t}^{\text{excess}} \equiv R_{j,t} - \alpha_t = \beta_{j,t} + \sum_{k=1}^K \omega_{j,k} \gamma_{k,t}. \quad (14)$$

The table shows that industry effects $\beta_{j,t}$ explain most of the excess variance of industry indices (which makes sense), while country-composition matters mainly when an industry is geographically concentrated.

Table 3
Decomposition of excess index returns into country and industry effects

The table gives the variance of the components of the equally-weighted (EW) and value-weighted (VW) excess country and industry index returns over the European market. Each excess country index return is decomposed in a pure country effect and a sum of seven industry effects. Each excess industry index return is decomposed in the sum of 12 country effects and a pure industry effect. Returns are measured in percent per month. The ratio relative to the market gives the ratio of the variance of that component to the variance of the index return in excess of the European market.

Country	EW indices				VW indices			
	Pure country effect ^a		Sum of 7 industry effects ^b		Pure country effect ^a		Sum of 7 industry effects ^b	
	Variance	Ratio relative to market	Variance	Ratio relative to market	Variance	Ratio relative to market	Variance	Ratio relative to market
Austria	34.29	1.016	0.08	0.002	15.74	1.028	0.76	0.022
Belgium	12.75	0.992	0.22	0.017	15.87	1.033	1.00	0.065
Denmark	23.97	0.997	0.19	0.008	26.23	1.030	0.83	0.033
France	18.93	0.998	0.04	0.002	24.33	0.989	0.10	0.004
Germany	10.61	1.027	0.05	0.005	14.01	0.987	0.56	0.019
Italy	35.09	1.002	0.03	0.001	44.26	0.994	0.63	0.014
Netherlands	15.58	1.001	0.09	0.006	7.14	0.754	4.06	0.429
Norway	35.18	0.984	0.18	0.005	40.00	0.874	3.80	0.083
Spain	44.73	1.083	0.48	0.012	33.99	1.053	1.99	0.062
Sweden	34.11	0.991	0.20	0.006	30.81	0.965	0.86	0.027
Switzerland	10.53	1.013	0.06	0.006	10.68	0.955	0.70	0.063
United Kingdom	14.35	0.997	0.02	0.001	8.74	0.967	0.08	0.009
Cross-country average	24.18	1.008	0.14	0.006	24.32	0.969	1.28	0.071

Industry	EW indices		VW indices		EW indices		VW indices	
	Sum of 12 country effects ^a		Pure industry effect ^a		Sum of 12 country effects ^a		Pure industry effect ^a	
	Variance	Ratio relative to market	Variance	Ratio relative to market	Variance	Ratio relative to market	Variance	Ratio relative to market
Finance	0.07	0.054	1.24	0.954	0.28	0.125	1.88	0.839
Energy	1.10	0.058	17.67	0.933	1.50	0.077	18.48	0.948
Utilities	3.78	0.413	9.20	1.004	4.84	0.441	8.14	0.741
Transport	1.81	0.243	5.89	0.792	0.90	0.087	9.66	0.937
Consumer	0.17	0.187	0.76	0.835	0.18	0.118	1.51	0.987
Capital	0.33	0.164	1.87	0.930	0.25	0.072	3.11	0.899
Basic	0.31	0.208	1.36	0.913	0.88	0.317	2.47	0.888
Cross-industry average	1.08	0.190	5.43	0.909	1.26	0.177	6.46	0.891

^aThe pure country effect measures the average return of firms in a country relative to firms which are in the same industry but located in a different country.

^bThe sum of the seven industry effects represents the component of a country's return that can be attributed to the difference between the industrial composition of its market and the industrial composition of the European market.

^cThe pure industry effect measures the average return of firms in an industry relative to firms which are located in the same country but belong to a different industry.

^dThe sum of the 12 country effects represents the component of an industry's return that can be attributed to the difference between the geographical composition of its index and the geographical composition of the European market.

5.4 Table 4: “Corrected” indices — controlling for industry does not remove country diversification gains

Table 4 repeats summary statistics and correlation matrices for:

- *Country* returns corrected for industry effects ($\tilde{R}_{k,t} = \alpha_t + \gamma_{k,t}$), and
- *Industry* returns corrected for country effects ($\tilde{R}_{j,t} = \alpha_t + \beta_{j,t}$).

Interpretation:

- Country correlations remain low even after removing industrial composition. So the “industrial structure explains international diversification” story is not supported.
- Industry correlations remain high after removing country composition. So industries share a big common component that is not mainly due to country weights.

Table 4
Summary statistics: Estimated country and industry indices
Panel A summarizes the mean and the standard deviation of the monthly equally-weighted (EW) and value-weighted (VW) estimated country returns, corrected for industry effects. Panel B contains the summary statistics for the monthly returns on equally- and value-weighted industry portfolios, corrected for country effects. All returns are measured in Deutschmarks and expressed in percent per month. The correlations above the diagonal refer to value-weighted index returns, those below the diagonal are between the equally-weighted index returns.

Country		EW return		VW return		Correlations															
		Mean		Std. dev.		A	B	D	F	G	I	Ne	No	Sp	Swe	Swi	UK	Eur			
		Mean	Std. dev.	Mean	Std. dev.																
Austria	(A)	0.813	6.571	0.713	6.450																
Belgium	(B)	1.300	5.357	1.155	5.271	0.376															
Denmark	(D)	1.099	5.480	0.988	5.390	0.147	0.389														
France	(F)	1.483	6.553	1.374	6.426	0.338	0.582	0.314													
Germany	(G)	0.950	4.651	0.932	5.276	0.539	0.616	0.395	0.527												
Italy	(I)	1.509	7.732	1.507	7.981	0.282	0.401	0.326	0.419	0.434											
Netherlands	(Ne)	1.213	5.515	1.317	4.654	0.306	0.601	0.397	0.488	0.599	0.324										
Norway	(No)	0.795	7.758	1.122	7.912	0.324	0.478	0.453	0.389	0.430	0.409	0.452									
Spain	(Sp)	1.222	8.688	1.102	7.309	0.280	0.471	0.361	0.406	0.399	0.409	0.446	0.422								
Sweden	(Swe)	1.648	7.283	1.436	6.909	0.236	0.377	0.329	0.315	0.356	0.377	0.463	0.461	0.425							
Switzerland	(Swi)	0.747	4.931	0.902	4.645	0.445	0.637	0.416	0.481	0.685	0.373	0.675	0.496	0.441	0.486						
United Kingdom	(UK)	1.434	6.388	1.336	6.000	0.305	0.516	0.401	0.428	0.462	0.400	0.502	0.508	0.501	0.465	0.570					
Europe	(Eur)	1.234	4.524	1.105	4.412	0.486	0.751	0.535	0.748	0.748	0.645	0.708	0.648	0.652	0.598	0.708	0.812				

Industry		EW return		VW return		Correlation matrix									
		Mean		Std. dev.		F	E	U	T	Co	Ca	B	Eur		
		Mean	Std. dev.	Mean	Std. dev.										
Finance	(F)	1.181	4.387	1.124	4.680										
Energy	(E)	1.485	5.578	1.278	6.059	0.625									
Utilities	(U)	1.345	3.765	1.158	4.290	0.731	0.531								
Transport	(T)	1.268	5.239	0.899	5.530	0.842	0.546	0.633							
Consumer	(Co)	1.360	4.459	1.248	4.535	0.940	0.636	0.713	0.856						
Capital	(Ca)	0.968	5.184	0.848	5.019	0.920	0.613	0.715	0.862	0.941					
Basic	(B)	1.204	5.171	0.986	4.916	0.928	0.652	0.696	0.877	0.945	0.951				
Europe	(Eur)	1.234	4.524	1.105	4.412	0.969	0.672	0.747	0.887	0.981	0.969	0.980			

^aThe corrected country index returns are computed as the sum of the return on the European equally- or value-weighted market and the estimated equally- or value-weighted country-specific component of return variation in each country, and therefore exclude the sources of return variation that are due to the industrial composition of the country indices.

^bThe corrected industry index returns are computed as the sum of the return on the European equally- or value-weighted market and the estimated equally- or value-weighted industry-specific component of return variation in each sector, and therefore exclude the sources of return variation that are due to the geographical composition of the industry indices.

5.5 Figure 1: Diversification is more powerful across countries than across industries

Figure 1 compares how fast portfolio variance falls as I add stocks when: (i) I diversify across industries within one country, (ii) I diversify across countries within one industry, (iii) I diversify across both.

Read:

- The within-country-across-industry curve bottoms out at a much higher residual variance.
- The within-industry-across-country curve drops much further.
- Adding both dimensions only marginally improves beyond country diversification.

Intuition: If country shocks are large and not spanned by industry shocks, then country diversification is the key margin.

5.6 Table 5: Why “industry factors from aggregate country returns” can look misleading

The paper revisits a common aggregate approach: regress country index returns on their industry weights to infer “industry factor returns.” Each month t , estimate

$$R_{k,t} = \sum_{j=1}^J \omega_{k,j} f_{j,t} + u_{k,t}, \quad (15)$$

(with no constant because $\sum_j \omega_{k,j} = 1$).

Interpretation of Table 5:

- The implied $f_{j,t}$ can have extremely large volatilities and strong negative correlations.
- This happens because when industry weights are similar across countries, the regression is close to singular. So the fitted “industry factor” portfolios become highly leveraged long–short combinations of country indices.
- Those $f_{j,t}$ are therefore not the same as the economically meaningful, investable Europe-wide industry portfolios $\alpha_t + \beta_{j,t}$ coming from the stock-level decomposition.

Table 5
Summary statistics of industry factors estimated from aggregate data

The table gives the means, standard deviations, and the correlation matrix of the estimated industry factors using Roll’s (1992) methodology. In each month the industry factor returns are estimated as the coefficients in a cross-sectional regression (excluding a constant term) of the value-weighted (VW) country index returns on the weights of each of the seven industries in the country indices. By running a cross-sectional regression for each month during the period 1978–1992, we obtain a time series of industry factor returns. All returns are expressed in percent per month.

Industry	Factor return		Correlation matrix						
	Mean	Std. dev.	F	E	U	T	Co	Ca	
Finance (F)	1.670	14.459							
Energy (E)	1.124	9.946	0.310						
Utilities (U)	1.391	17.762	−0.427	−0.137					
Transportation (T)	−6.604	124.333	−0.393	−0.397	0.137				
Consumer (Co)	0.578	14.978	−0.538	−0.012	0.408	−0.122			
Capital (Ca)	4.318	30.673	−0.454	0.030	0.207	0.016	−0.523		
Basic (B)	−0.841	19.799	−0.342	0.103	−0.257	0.179	0.094	−0.624	

Table 6

Country effects and excess exchange rate returns

Slope and the R^2 of a regression of the country effect relative to the European equally-weighted (EW) and value-weighted (VW) index on the difference between the relative change in the DM/FX exchange rate and the relative change in the exchange rate of the DM versus the weighted average of European currencies (EW and VW). Standard errors are in parentheses.

Country	EW		VW	
	Slope	R^2	Slope	R^2
Austria	0.591 (0.400)	0.012	0.542 (0.333)	0.015
Belgium	0.432 (0.258)	0.016	0.661 (0.234)	0.043
Denmark	0.980 (0.275)	0.067	0.713 (0.248)	0.044
France	1.804 (0.285)	0.184	1.486 (0.253)	0.163
Germany	1.377 (0.230)	0.168	1.266 (0.211)	0.168
Italy	0.960 (0.238)	0.046	0.746 (0.316)	0.030
Netherlands	0.831 (0.310)	0.039	0.664 (0.167)	0.081
Norway	1.036 (0.315)	0.057	1.057 (0.309)	0.061
Spain	1.591 (0.340)	0.110	1.336 (0.287)	0.109
Sweden	0.309 (0.258)	0.008	0.374 (0.231)	0.014
Switzerland	0.849 (0.123)	0.212	0.846 (0.117)	0.227
United Kingdom	0.991 (0.135)	0.232	0.971 (0.126)	0.249

5.7 Table 6: Country effects are not mainly currency effects

The paper constructs “excess” exchange rate returns (relative to a European basket) and tests whether they explain estimated country effects. Let $S_{k,t}$ be the exchange rate return for country k

(common-currency return minus local-currency return). Define the relative currency move:

$$X_{k,t}^{\text{EW}} = S_{k,t} - \sum_{m=1}^K \pi_m^{\text{EW}} S_{m,t}, \quad \pi_m^{\text{EW}} = \frac{m_m}{\sum_{\ell} m_{\ell}}, \quad (16)$$

$$X_{k,t}^{\text{VW}} = S_{k,t} - \sum_{m=1}^K \pi_m^{\text{VW}} S_{m,t}, \quad \pi_m^{\text{VW}} = v_m. \quad (17)$$

Then estimate (for each country k):

$$\gamma_{k,t} = a_k + b_k X_{k,t} + e_{k,t}. \quad (18)$$

Interpretation: R^2 values are generally low. Currency moves explain only a modest fraction of the country effects. So the big “country” component is not simply exchange-rate risk.

6 Short summary

- A clean empirical decomposition of stock returns into European market, country, and industry components using monthly cross-sectional regressions.
- In Europe (1978–1992), **country effects dominate**: industrial specialization explains little of the variation in excess country index returns.
- Therefore, **international diversification benefits are not mainly industry diversification in disguise**.
- Exchange rates do not account for the large country effects.

7 Conclusion

7.1 Summary

- **Research question.** The paper asks whether the gains from international diversification are mainly an *industry-composition* story: if countries are just different portfolios of the same industries, then “country diversification” would be almost equivalent to “industry diversification.”
- **Methodological contribution.** Each month, the paper runs a two-way return decomposition at the stock level, separating a *European market component*, *industry effects*, and *country effects*. This produces economically interpretable “pure” country and industry return series (rather than mechanically extracting industry factors from aggregate country indices, which can be ill-conditioned).
- **Main empirical conclusion.** Differences in industrial structure across countries explain *very little* of the variation in excess country returns. In the variance decomposition (Table

3), the *cross-country average* share of the excess country-index variance attributable to the *industry-mix* component is only about

$$0.006 \text{ (equally-weighted)} \quad \text{and} \quad 0.071 \text{ (value-weighted)},$$

implying that *most* excess country variation is driven by **pure country effects**.

- **Implication for diversification.** Since most of the non-European-market variation in country indices is country-specific, international diversification remains powerful even if an investor already diversifies across industries.
- **Why the “industrial structure explains it” hypothesis fails empirically.** When the paper constructs country indices *corrected* for industry composition (Table 4), cross-country correlations remain low. Thus the diversification gains do not disappear after controlling for industry mix.
- **Currency is not the dominant explanation either.** Regressions relating estimated country effects to exchange-rate movements (Table 6) yield generally low explanatory power, so the country component is not mainly a currency component.
- **Portfolio-level evidence.** The diversification frontier in Figure 1 shows that reducing risk by adding *countries* is more effective than adding *industries* (within Europe in this sample), consistent with large country-specific shocks.

7.2 Economic intuition: why country diversification works beyond industries

- **Country shocks are not spanned by industry shocks.** Even if firms belong to the same industry, their cash flows and discount rates can co-move because they share country-level forces: monetary/fiscal policy, institutions, political risk, local credit conditions, labor-market regimes, and country-specific business cycles.
- **Industry portfolios within Europe share a strong common component.** Europe-wide industry returns are highly correlated because many industries load on a common regional market factor. This means “industry diversification” alone cannot eliminate country-level residual risk.
- **Why aggregate regressions on industry weights can mislead.** If most countries hold broadly similar industry weights, then regressing country index returns on those weights is close to singular. The implied “industry factors” can become extreme long–short combinations (high volatility, unstable correlations), so they are not the economically meaningful industry portfolios that a real investor would hold.

7.3 Takeaway

Industrial composition differences across European countries are real, but they explain only a small fraction of the time-series variation in excess country returns. A large country-specific component remains after stripping out industry effects, and it is not

mainly currency risk. Therefore, the benefits of international diversification are not simply industry diversification in disguise: country factors matter.